FROM MASSES TO AGES

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ABSTRACT

FROM MASSES TO AGES

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CHAPTER 1

Introduction

1.1. Formation Mechanisms

1.1.1. Core Accretion

The widely-accepted model for core accretion (CA hereon) comes from Pollack et al., 1996. A general overview of the process is as follows: a protoplanetary disk consists of dust and gas. This dust undergoes random motion, and eventually dust particles may collide and clump together. The repetition of this process eventually leads to pebbles (dust coagulation, Drążkowska et al., 2023), then boulders, plentesimals, then finally, through planetesimal accretion (Safronov and Zvjagina, 1969), a planetary core massive enough to start accreting gas from the disk, which eventually leads to runaway gas accretion (Mizuno, 1980, Bodenheimer and Pollack, 1986, Pollack et al., 1996). All in all, this process' timescale is on the scale of Myr (Pollack et al., 1996), which is significant because a planet formed in this way would be sizeably younger than its host star; planets formed through CA are also what this project aims to find.

This process occurs closer to the centre of the disk than gravitational instability, the other widely-accepted formation pathway, detailed in the next section. Note that this process is not necessarily linear; most pebbles actually don't evolve to the metre-scale. Among the potential barriers include the fragmentation-barrier, drift-barrier and others that are beyond the scope of this project but are detailed in Drazkowska et al. 2023.

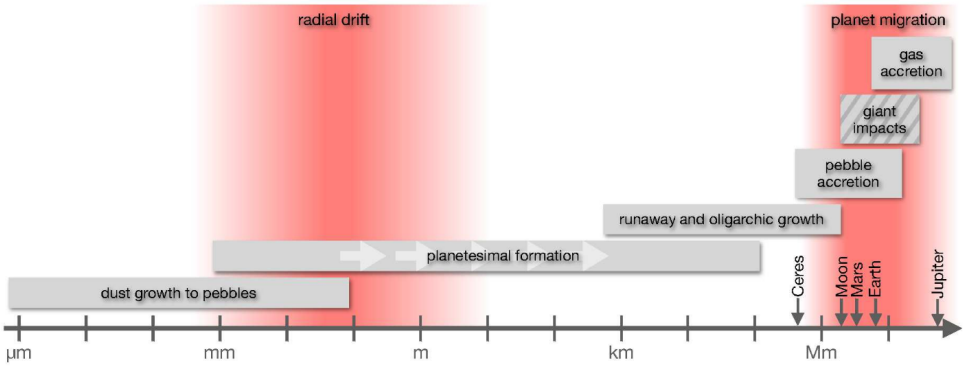


Figure 1.1: Fig. 3 from Drazkowska2023. This figure provides a general scale for the stages of CA.

Previous models that Pollack et al., 1996 model built upon include Mizuno et al., 1978 and Mizuno, 1980 of core instability, Bodenheimer and Pollack, 1986's investigations into Mizuno's models, Pollack et al., 1986 and Podolak et al., 1988 examinations into planetesimal accretion, among others.

In short, a critical mass before runaway gas accretion occurs is found to be $\sim 10M_{\oplus}$ (Mizuno, 1980), later $\sim 10M_{\oplus} - 30M_{\oplus}$ (BP86), and also when the heavy-element mass and the gas mass of the planet would equal; after this critical mass is reached, runaway gas accretion occurs. However, observations of the giant planets also revealed that Neptune and Uranus had compositions such that, in these models, their mass fractions were comprised of too little gas mass, so they'd have never reached the equality required to reach the critical mass. Pollack et al., 1996's remedy to this issue will be found in the outline below.

Arranged neatly, the model of Pollack et al., 1996 is split into three phases: during phase 1, planetesimal accretion exceeds gas accretion rate; they're initially many orders of magnitude apart, with planetesimal accretion rate being the greater value. Overtime, the two rates become comparable: after $\sim 10^5$ years, they are comparable and planetesimal accretion rate begins declining sharply. Meanwhile, gas accretion rate keeps growing. This phase denotes a runaway planetesimal accretion, separate from the runaway gas accretion, and it finishes when a protoplanet has emptied its feeding zone of planetesimals. However, the feeding zone grows as the planet's total mass increases,

introducing new planetesimals (Pollack et al., 1996)

Phase 2 is a novel phase introduced in this model, that denotes a long, liminal phase where both accretion rates are relatively time-invariant, now with gas accretion rate exceeding planetesimal accretion rate. This phase spans the bulk of the aforementioned Myr timescale, with Pollack's simulations of Jupiter lasting $\sim 7Myr$. Finally, phase 3 denotes the runaway gas accretion, where both accretion rates (with gas still far exceeding planetesimal) start rapidly increasing again.

Fig 1c of Pollack et al., 1996 is particluarly relevant to this project: during phase 3, the luminosity grows rapidly to a stage that we can image directly. Once it reaches a maximum luminosity, it will cool down for the rest of its lifetime (since planets have no mechanisms that illuminate them); this cooling is what this project will utilise to infer the age from the luminosity and dynamical masses.

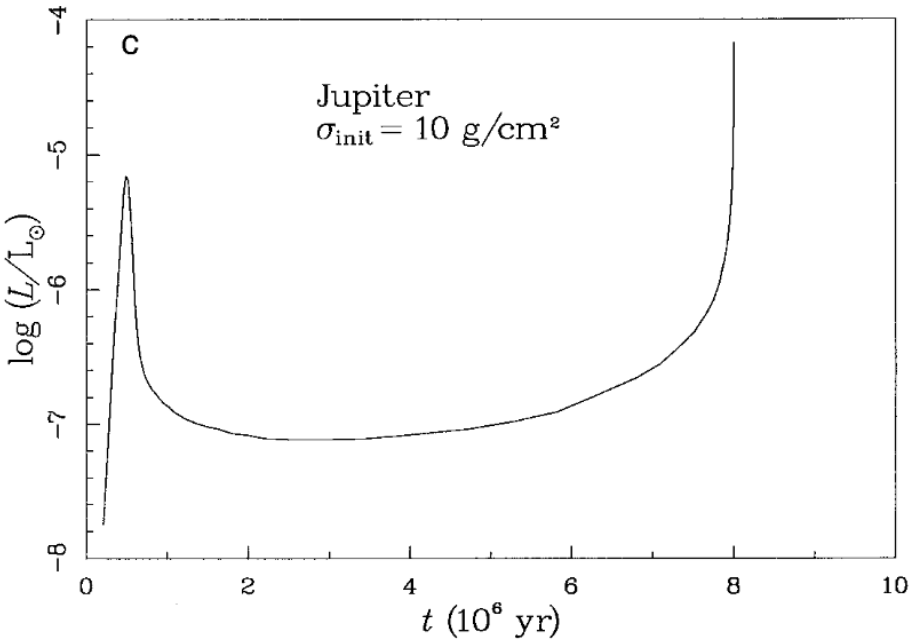


Figure 1.2: Fig 1c of Pollack et al., 1996. The initial rise corresponds to the runaway planetesimal accretion of phase 1; the long plateau corresponds to the transitionary phase 2; the sharp increasing corresponds to both accretion rates rapidly increasing (and the runaway gas accretion) in phase 3

1.2. Getting Started

The template requires TeX Live 2022 and is optimized for pdfLaTeX and the LaTeX editor Overleaf. Other LaTeX editing programs may require you to compile the document more than once. To claim your free Overleaf Pro account, go to the Penn Libraries Overleaf Portal and click on the green "Log in through your institution" button.

Click here if you have a question about Overleaf, and they'll make sure it reaches the right person. Click here for their help page to learn how to use Overleaf and LaTeX. See Chapter 1.5 for additional resources.

1.2.1. Packages

The template automatically loads the packages it requires to function. See the PackagesLoadedbyTemplate.tex file for a complete list of all the packages loaded by the template. You may manually load other packages if you wish, and Overleaf will give you a warning message when you compile your document if your package conflicts with one of the template's packages. If you receive a warning, you can consult the specified package documentation on CTAN for a possible resolution.

1.3. Margins

As of January 2023, the margins must be 1 inch on each side. Remember to resolve all overfull or underfull warnings to help ensure your dissertation meets the margin requirements.[1] Click here to learn how to identify and resolve the warnings. Common causes of warnings include

- **LaTeX CANNOT automatically break long mathematical expressions or parenthetical references across multiple lines.** Chapter 3 demonstrates two methods for manually splitting equations.
- You can manually control the hyphenation by using the `\hyphenation{}` hints command. For example, write `\hyphenation{rep-re-sent-a-tive}` in the preamble to tell LaTeX where it can break the word "representative."
- The Verbatim environment and the `\Verb` command remove LaTeX's formatting controls. There may be instances that require you to manually adjust line breaks.
- The breaklinks option of the hyperref package is activated, however there may be instances of URLs running into the margins that require your manual intervention.

1.4. Template Tips

1.4.1. Chapters, sections, and subsections

Create chapters with the `\chapter` command, sections with the `\section` command, and subsections with the `\subsection` command. If you use the `\MakeUppercase` command to autocapitalize chapter titles, carefully check each title and its table of contents entry for compliance with University formatting requirements. You may wish to split chapters into separate .tex files for easier document management. You can use the `\include{}` command to add the chapters to your main .tex file. Click here to learn more.

If a chapter or section title contains math expressions or other symbols, the `hyperref` package may give a PDFDocEncoded warning when you compile your document. Use the `\texorpdfstring`

[1] This is a common source of errors – please double check the margins before submitting your final dissertation.

command to edit the corresponding PDF bookmark text to resolve the warning. For example, the section title "Lemurs Wrote $E = mc^2$ in Mud on the Riverbank" should be coded

```
\section{Lemurs Wrote \texorpdfstring{$E=mc^2$}{E=mc2} in Mud on the Riverbank}
```

1.4.2. Page orientation

The template's default page orientation is portrait. Use the landscape environment to create landscape pages. Page number location and margins will automatically update like page 9.

1.5. Getting Help and Giving Feedback

Questions about LaTeX?

- Click here to connect with LaTeX resources, tutorials, and more.

Want to give feedback on the template?

- Click here to share it with us.

1.6. Acknowledgments

The University of Pennsylvania Ph.D. Dissertation LaTeX Template was initiated by Dr. Beth S. Wenger, Associate Dean for Graduate Studies, School of Arts and Sciences. It was developed by Lauren Gala, Penn Libraries with assistance from Dr. Julia Hartmann and Dr. Man Cheung Tsui, Department of Mathematics, and Bridget Rothera, Graduate Division School of Arts and Sciences. Portions of the code and text were reprinted or adapted with permission from Ratcliffe S.J. (2017) Penn Biostat LaTeX Template: PhD dissertation modified from Dissertation Template for Wharton PhD Candidates in LaTeX.

CHAPTER 2

TABLES AND FIGURES

2.1. Tables

The captions of tables inside the table environment (see Table 2.1) will display in the List of Tables. Use the `\label` to label tables.

		Primes				
		2	3	5	7	
Powers	504	3	2	0	1	
	540	2	3	1	0	
Powers	gcd	2	2	0	0	min
	lcm	3	3	1	1	max

Table 2.1: Example Table

2.2. Figures

The captions of figures inside the figure environment (see Figure 2.1) will display in the List of Illustrations. Use the `\label` to label figures.[2]

[2]This also includes photographs, maps, schemes, diagrams, etc.



Figure 2.1: Lemur photograph

LaTeX takes into account line and page breaks when determining where to place floating objects like figures and tables on a page. You can override LaTeX's placement decisions by using the float package options. Consult the float package documentation to learn how to gain more control over where and how float objects display.

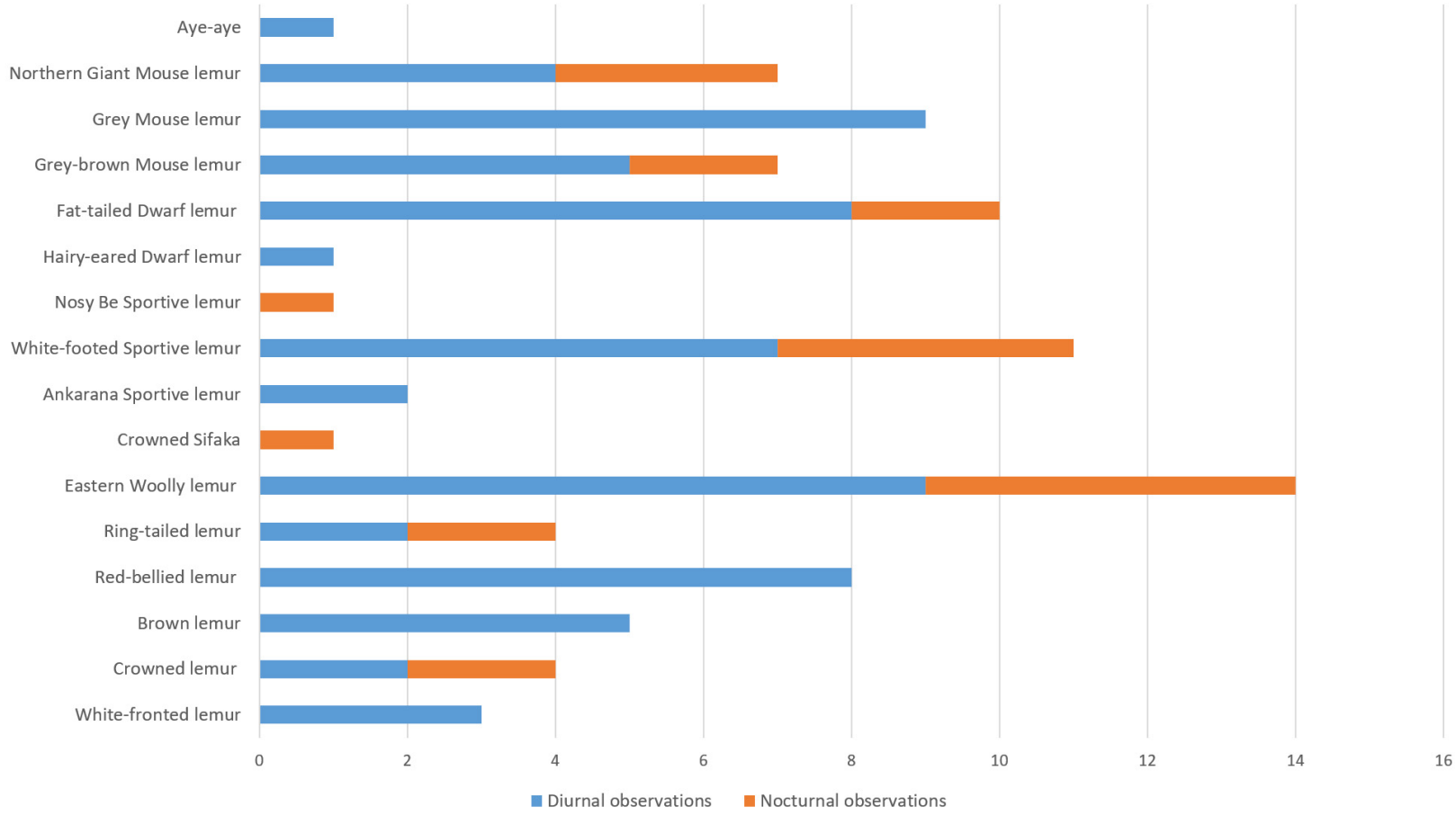


Figure 2.2: Lemur bar chart on landscape page

CHAPTER 3

EQUATIONS

The template automatically loads several math packages like amsmath, amssymb, and amsfonts. The split and the multline environments allow you to split an equation over multiple lines. Equation 3.1 is written in the split environment, and Equation 3.2 is in the multline environment. Notice the differences in how the two environments align the equations and where they place the equation number.

$$\begin{aligned} A &= \frac{\pi r^2}{2} \\ &= \frac{1}{2}\pi r^2 \end{aligned} \tag{3.1}$$

$$\begin{aligned} p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 + 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 \\ + 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3 \end{aligned} \tag{3.2}$$

CHAPTER 4

APPENDICES

Appendices are optional. Use the `\chapter` command inside the custom `append` environment to insert appendices. Use the `\ref{ }` command to reference appendices in your text.

Tables and figures in appendices display in the List of Tables and the List of Illustrations, respectively. To prevent appendix tables and figures from displaying in either list, insert an asterisk between `\caption` and `{ }`.

CHAPTER 5

CITATIONS AND BIBLIOGRAPHY

LaTeX bibliography packages and citation management programs enable you to efficiently cite and format bibliographic references. You can manage bibliographic entries in a separate bibliography database file (*.bib) or you can embed the entries directly into the main .tex file. While the LaTeX template was built for the former approach, you can modify it to accommodate latter.

5.1. Getting Started

Please watch this short video to learn the principles underpinning the template bibliography before proceeding.

5.2. Default Template Settings

The commands written in sampledissertation.tex load the natbib package in its preamble and reference the bibliographic entries stored in bibliography.bib.

This is an example inline citation **?**. If you want to cite two publications in one place like **??**, write one citation command. You can add a prenote or postnote **?**, p. 78 to a citation.

5.3. Customization

Citation and bibliography formatting commands are written in the main .tex file for your convenience. You can edit the relevant code in the .tex file to produce citation and bibliography styles appropriate for your field. If you wish to make significant to changes to the structure of the bibliography, you may need to edit the bibliof environment defined in upennstyle.sty.

5.3.1. Styles and formatting

The natbib package, selected format and style options, and example citations are for demonstration. In addition to natbib, other packages and accompanying documentation are free to download from CTAN. Always consult the package documentation for additional citation and bibliography commands and formatting options. Remember that the available commands and styles differ between packages.

5.3.2. Embedded bibliography

You can edit the template to manage your bibliography directly in the .tex file obviating the need for a separate .bib file. Replace the `\bibliography` command in the bibliof environment with the list of bibitems and edit relevant code elsewhere in the template as needed.

5.4. Recommended Resources

- Click here to learn how to manage citations and bibliographies in LaTeX documents
- Click here to learn citation best practices and how to use citation management programs
- Click here to watch a short introduction to citing and formatting references with natbib

APPENDIX A

TITLE OF APPENDIX A

The content of Appendix A begins here. Use the `\chapter` command to insert additional appendices.

A.1. Section Name of Appendix A

Use the `\section` command to create sections.

APPENDIX B

TITLE OF APPENDIX B

The content of Appendix B begins here.

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